ENGR 421/DASC 521: Introduction to Machine Learning
Fall 2021 Final – Solution Key

Question 1:
We have two classes that are assumed to have one-dimensional Gaussian distributions with different means and variances: $p(x|y=1) \sim N(\mu_1, \sigma^2)$ and $p(x|y=2) \sim N(\mu_2, 4\sigma^2)$. Derive the position(s) of the intersection of the two posterior probabilities.

We would like to find x that satisfy $P(y=1|x) = P(y=2|x)$.

$$p(x|y=1)P(y=1) = p(x|y=2)P(y=2)$$
$$\log p(x|y=1) + \log P(y=1) = \log p(x|y=2) + \log P(y=2)$$
$$-\frac{\log(2\pi\sigma^2)}{2} - \frac{(x-\mu_1)^2}{2\sigma^2} + \log P(y=1) = -\frac{\log(8\pi\sigma^2)}{2} - \frac{(x-\mu_2)^2}{8\sigma^2} + \log P(y=2)$$

$$-\frac{3}{8\sigma^2}x^2 + \frac{4\mu_1 - \mu_2}{4\sigma^2}x - \frac{(2\mu_1 - \mu_2)(2\mu_1 + \mu_2)}{8\sigma^2} + \log\frac{P(y=1)}{P(y=2)} + \log 2 = 0$$

Question 2:
Consider a data set of N data points, in which each data point has one real-valued positive input x_i and the corresponding real-valued output y_i, i.e., $\{(x_i, y_i)\}_{i=1}^N$. We use the following model to fit the data, which has an unknown parameter w (the variance is known in advance and is set to 1).

$$p(y_i|x_i) \sim N(\log(wx_i), 1) \quad \forall i$$

(a) Describe a maximum likelihood approach to infer w and write down the log-likelihood objective for this problem.

(b) Find the maximum likelihood solution for w.

(a) By assuming the data points are independent from each other, we can write down the likelihood function as follows:

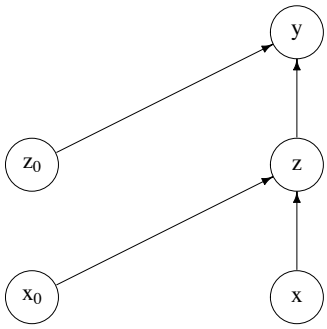
$$\text{likelihood} = \prod_{i=1}^N p(y_i|x_i) = \prod_{i=1}^N N(\log(wx_i), 1)$$
$$= \prod_{i=1}^N \frac{1}{\sqrt{2\pi(1)^2}} \exp\left(-\frac{(y_i - \log(wx_i))^2}{2(1)^2}\right)$$

$$\text{log-likelihood} = \sum_{i=1}^N \left(-(1/2)\log(2\pi) - (1/2)(y_i - \log(wx_i))^2\right)$$

(b) To maximize log-likelihood, we need to minimize $\sum_{i=1}^{N}(1/2)(y_i - \log(wx_i))^2$ with respect to w.

$$\frac{\partial \sum_{i=1}^{N}(1/2)(y_i - \log(wx_i))^2}{\partial w} = \sum_{i=1}^{N} \frac{\partial (1/2)(y_i - \log(wx_i))^2}{\partial w}$$
$$= \sum_{i=1}^{N}(2(1/2)(y_i - \log(wx_i)))(-(x_i/(wx_i))) = 0$$
$$\sum_{i=1}^{N}(y_i - (\log(w^\star) + \log(x_i))) = 0 \Rightarrow N\log(w^\star) = \sum_{i=1}^{N}(y_i - \log(x_i))$$
$$w^\star = \exp\left(\left[\sum_{i=1}^{N}(y_i - \log(x_i))\right]/N\right)$$

Question 3:
Given a multilayer perceptron with one input, one tanh hidden unit, and one sigmoid output unit, derive the weight update equations to minimize the cross-entropy using gradient-descent.



$$\frac{\partial \tanh(a)}{\partial a} = (1 - \tanh(a))^2$$
$$\frac{\partial \text{sigmoid}(a)}{\partial a} = \text{sigmoid}(a)(1 - \text{sigmoid}(a))$$
$$z_i = \tanh(wx_i + w_0)$$
$$\widehat{y}_i = \text{sigmoid}(vz_i + v_0)$$
$$\text{Error}_i = -y_i \log \widehat{y}_i - (1 - y_i)\log(1 - \widehat{y}_i)$$

$$\Delta v = -\eta \frac{\partial \text{Error}_i}{\partial \widehat{y}_i}\frac{\partial \widehat{y}_i}{\partial v}$$
$$= \eta(y_i - \widehat{y}_i)z_i$$
$$\Delta v_0 = -\eta \frac{\partial \text{Error}_i}{\partial \widehat{y}_i}\frac{\partial \widehat{y}_i}{\partial v_0}$$
$$= \eta(y_i - \widehat{y}_i)$$
$$\Delta w = -\eta \frac{\partial \text{Error}_i}{\partial \widehat{y}_i}\frac{\partial \widehat{y}_i}{\partial z_i}\frac{\partial z_i}{\partial w}$$
$$= \eta(y_i - \widehat{y}_i)v(1 - z_i)^2 x_i$$
$$\Delta w_0 = -\eta \frac{\partial \text{Error}_i}{\partial \widehat{y}_i}\frac{\partial \widehat{y}_i}{\partial z_i}\frac{\partial z_i}{\partial w_0}$$
$$= \eta(y_i - \widehat{y}_i)v(1 - z_i)^2$$

Question 4:
We know that an $N \times N$ symmetric real matrix $\mathbf{K}$ is said to be positive semidefinite if $\boldsymbol{a}^\top \mathbf{K} \boldsymbol{a} \geq 0$ for all $\boldsymbol{a}$ in $\mathbb{R}^N$.

(a) Show that $k(\boldsymbol{x}_i, \boldsymbol{x}_j) = \boldsymbol{x}_i^\top \boldsymbol{x}_j$ produces a positive semidefinite kernel matrix on a given set of N data points, i.e., $\mathcal{X} = \{\boldsymbol{x}_i \in \mathbb{R}^D\}_{i=1}^N$.

(b) Show that $k(\boldsymbol{x}_i, \boldsymbol{x}_j) = \dfrac{\boldsymbol{x}_i^\top \boldsymbol{x}_j}{\|\boldsymbol{x}_i\|_2 \|\boldsymbol{x}_j\|_2}$ produces a positive semidefinite kernel matrix on a given set of N data points, i.e., $\mathcal{X} = \{\boldsymbol{x}_i \in \mathbb{R}^D\}_{i=1}^N$.

(a) $k(\boldsymbol{x}_i, \boldsymbol{x}_j) = \boldsymbol{x}_i^\top \boldsymbol{x}_j$ produces the following kernel matrix:

$$\mathbf{K} = \begin{bmatrix} \boldsymbol{x}_1^\top \boldsymbol{x}_1 & \boldsymbol{x}_1^\top \boldsymbol{x}_2 & \dots & \boldsymbol{x}_1^\top \boldsymbol{x}_N \\ \boldsymbol{x}_2^\top \boldsymbol{x}_1 & \boldsymbol{x}_2^\top \boldsymbol{x}_2 & \dots & \boldsymbol{x}_2^\top \boldsymbol{x}_N \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{x}_N^\top \boldsymbol{x}_1 & \boldsymbol{x}_N^\top \boldsymbol{x}_2 & \dots & \boldsymbol{x}_N^\top \boldsymbol{x}_N \end{bmatrix} = \underbrace{\begin{bmatrix} \boldsymbol{x}_1^\top \\ \boldsymbol{x}_2^\top \\ \vdots \\ \boldsymbol{x}_N^\top \end{bmatrix}}_{\mathbf{X}} \underbrace{\begin{bmatrix} \boldsymbol{x}_1^\top \\ \boldsymbol{x}_2^\top \\ \vdots \\ \boldsymbol{x}_N^\top \end{bmatrix}^\top}_{\mathbf{X}^\top}$$

$$\begin{aligned} \boldsymbol{a}^\top \mathbf{K} \boldsymbol{a} &\geq 0 \\ \boldsymbol{a}^\top \mathbf{X}\mathbf{X}^\top \boldsymbol{a} &\geq 0 \\ (\mathbf{X}^\top \boldsymbol{a})^\top \mathbf{X}^\top \boldsymbol{a} &\geq 0 \\ \|\mathbf{X}^\top \boldsymbol{a}\|_2^2 &\geq 0 \end{aligned}$$

(b) $k(\boldsymbol{x}_i, \boldsymbol{x}_j) = \dfrac{\boldsymbol{x}_i^\top \boldsymbol{x}_j}{\|\boldsymbol{x}_i\|_2 \|\boldsymbol{x}_j\|_2}$ produces the following kernel matrix:

$$\mathbf{K} = \begin{bmatrix} \dfrac{\boldsymbol{x}_1^\top \boldsymbol{x}_1}{\|\boldsymbol{x}_1\|_2 \|\boldsymbol{x}_1\|_2} & \dfrac{\boldsymbol{x}_1^\top \boldsymbol{x}_2}{\|\boldsymbol{x}_1\|_2 \|\boldsymbol{x}_2\|_2} & \cdots & \dfrac{\boldsymbol{x}_1^\top \boldsymbol{x}_N}{\|\boldsymbol{x}_1\|_2 \|\boldsymbol{x}_N\|_2} \\ \dfrac{\boldsymbol{x}_2^\top \boldsymbol{x}_1}{\|\boldsymbol{x}_2\|_2 \|\boldsymbol{x}_1\|_2} & \dfrac{\boldsymbol{x}_2^\top \boldsymbol{x}_2}{\|\boldsymbol{x}_2\|_2 \|\boldsymbol{x}_2\|_2} & \cdots & \dfrac{\boldsymbol{x}_2^\top \boldsymbol{x}_N}{\|\boldsymbol{x}_2\|_2 \|\boldsymbol{x}_N\|_2} \\ \vdots & \vdots & \ddots & \vdots \\ \dfrac{\boldsymbol{x}_N^\top \boldsymbol{x}_1}{\|\boldsymbol{x}_N\|_2 \|\boldsymbol{x}_1\|_2} & \dfrac{\boldsymbol{x}_N^\top \boldsymbol{x}_2}{\|\boldsymbol{x}_N\|_2 \|\boldsymbol{x}_2\|_2} & \cdots & \dfrac{\boldsymbol{x}_N^\top \boldsymbol{x}_N}{\|\boldsymbol{x}_N\|_2 \|\boldsymbol{x}_N\|_2} \end{bmatrix} = \underbrace{\begin{bmatrix} \dfrac{\boldsymbol{x}_1^\top}{\|\boldsymbol{x}_1\|_2} \\ \dfrac{\boldsymbol{x}_2^\top}{\|\boldsymbol{x}_2\|_2} \\ \vdots \\ \dfrac{\boldsymbol{x}_N^\top}{\|\boldsymbol{x}_N\|_2} \end{bmatrix}}_{\mathbf{U}} \underbrace{\begin{bmatrix} \dfrac{\boldsymbol{x}_1^\top}{\|\boldsymbol{x}_1\|_2} \\ \dfrac{\boldsymbol{x}_2^\top}{\|\boldsymbol{x}_2\|_2} \\ \vdots \\ \dfrac{\boldsymbol{x}_N^\top}{\|\boldsymbol{x}_N\|_2} \end{bmatrix}^\top}_{\mathbf{U}^\top}$$

$$\begin{aligned} \boldsymbol{a}^\top \mathbf{K} \boldsymbol{a} &\geq 0 \\ \boldsymbol{a}^\top \mathbf{U}\mathbf{U}^\top \boldsymbol{a} &\geq 0 \\ (\mathbf{U}^\top \boldsymbol{a})^\top \mathbf{U}^\top \boldsymbol{a} &\geq 0 \\ \|\mathbf{U}^\top \boldsymbol{a}\|_2^2 &\geq 0 \end{aligned}$$

Question 5:
The XOR problem is given by:

$\boldsymbol{x}_i$	x_{i1}	x_{i2}	y_i
$\boldsymbol{x}_1$	-1	-1	-1
$\boldsymbol{x}_2$	$+1$	-1	$+1$
$\boldsymbol{x}_3$	-1	$+1$	$+1$
$\boldsymbol{x}_4$	$+1$	$+1$	-1

If we use the following mapping function $\Phi(\cdot)$ for the support vector machine formulation, can we solve this binary classification problem successfully? Justify your answer.

$$\Phi(\boldsymbol{x}_i) = \begin{bmatrix} x_{i1}^2 - x_{i2}^2 \\ x_{i1}x_{i2} \\ x_{i1}^2 + x_{i2}^2 \end{bmatrix}$$

The mapping function produces the following representations: $\Phi(\boldsymbol{x}_1) = \begin{bmatrix} 0 & 1 & 2 \end{bmatrix}^\top$, $\Phi(\boldsymbol{x}_2) = \begin{bmatrix} 0 & -1 & 2 \end{bmatrix}^\top$, $\Phi(\boldsymbol{x}_3) = \begin{bmatrix} 0 & -1 & 2 \end{bmatrix}^\top$, and $\Phi(\boldsymbol{x}_4) = \begin{bmatrix} 0 & 1 & 2 \end{bmatrix}^\top$.

The kernel matrix can be constructed as follows:

$$\mathbf{K} = \begin{bmatrix} 5 & 3 & 3 & 5 \\ 3 & 5 & 5 & 3 \\ 3 & 5 & 5 & 3 \\ 5 & 3 & 3 & 5 \end{bmatrix}$$

$$\begin{aligned} \text{maximize} \quad & J = \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 - \frac{1}{2} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \end{bmatrix}^\top \begin{bmatrix} 5 & -3 & -3 & 5 \\ -3 & 5 & 5 & -3 \\ -3 & 5 & 5 & -3 \\ 5 & -3 & -3 & 5 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \end{bmatrix} \\ \text{with respect to} \quad & \alpha_1,\ \alpha_2,\ \alpha_3,\ \alpha_4 \\ \text{subject to} \quad & -\alpha_1 + \alpha_2 + \alpha_3 - \alpha_4 = 0 \\ & \alpha_i \geq 0 \quad i \in \{1, 2, 3, 4\} \end{aligned}$$

$$\begin{aligned} \frac{\partial J}{\partial \alpha_1} = \frac{\partial J}{\partial \alpha_4} &= 1 - 5\alpha_1 + 3\alpha_2 + 3\alpha_3 - 5\alpha_4 = 0 \\ \frac{\partial J}{\partial \alpha_2} = \frac{\partial J}{\partial \alpha_3} &= 1 + 3\alpha_1 - 5\alpha_2 - 5\alpha_3 + 3\alpha_4 = 0 \end{aligned}$$

For example, $\alpha_1^\star = 1/2$, $\alpha_2^\star = 1/2$, $\alpha_3^\star = 0$, and $\alpha_4^\star = 0$ is one of the optimum solutions.

Question 6:
Recall the error function for k-means clustering with K clusters, data points $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N$, and centers $\widehat{\boldsymbol{\mu}}_1, \ldots, \widehat{\boldsymbol{\mu}}_K$:

$$E = \sum_{i=1}^{N} \sum_{k=1}^{K} b_{ik} \|\boldsymbol{x}_i - \widehat{\boldsymbol{\mu}}_k\|_2^2$$

where b_{ik} is equal to 1 if data point $\boldsymbol{x}_i$ is closer to center $\widehat{\boldsymbol{\mu}}_k$ than to any other center and to 0 otherwise.

(a) Instead of updating $\{\widehat{\boldsymbol{\mu}}_k\}_{k=1}^K$ by computing the means, let us minimize E with batch gradient descent while holding $\{b_{ik}\}_{i=1,k=1}^{N,K}$ fixed. Derive the update formula for $\widehat{\boldsymbol{\mu}}_1$ with learning rate η.

(b) Derive the update formula for $\widehat{\boldsymbol{\mu}}_1$ with gradient descent on a single data point $\boldsymbol{x}_i$. Use learning rate η.

(c) Recall that in the update step of the standard algorithm, we assign each cluster center to the mean of the data points closest to that center. It turns out that a particular choice of the learning rate η, which may be different for each cluster, makes the two algorithms (batch gradient descent and the standard k-means algorithm) have identical update steps. Let us focus on the update for the first cluster, with center $\widehat{\boldsymbol{\mu}}_1$. Calculate the value of η so that both algorithms perform the same update for $\widehat{\boldsymbol{\mu}}_1$.

(a)

$$\frac{\partial E}{\partial \widehat{\boldsymbol{\mu}}_1} = \frac{\partial\left(\sum_{i=1}^{N} b_{i1}\|\boldsymbol{x}_i - \widehat{\boldsymbol{\mu}}_1\|_2^2 + \text{constant}\right)}{\partial \widehat{\boldsymbol{\mu}}_1} = -2\sum_{i=1}^{N} b_{i1}(\boldsymbol{x}_i - \widehat{\boldsymbol{\mu}}_1)$$

Therefore the update formula is $\widehat{\boldsymbol{\mu}}_1^{(t+1)} = \widehat{\boldsymbol{\mu}}_1^{(t)} + 2\eta\sum_{i=1}^{N} b_{i1}(\boldsymbol{x}_i - \widehat{\boldsymbol{\mu}}_1^{(t)})$

(b)

$$\frac{\partial E_i}{\partial \widehat{\boldsymbol{\mu}}_1} = \frac{\partial(b_{i1}\|\boldsymbol{x}_i - \widehat{\boldsymbol{\mu}}_1\|_2^2 + \text{constant})}{\partial \widehat{\boldsymbol{\mu}}_1} = -2b_{i1}(\boldsymbol{x}_i - \widehat{\boldsymbol{\mu}}_1)$$

Therefore the update formula is $\widehat{\boldsymbol{\mu}}_1^{(t+1)} = \widehat{\boldsymbol{\mu}}_1^{(t)} + 2\eta b_{i1}(\boldsymbol{x}_i - \widehat{\boldsymbol{\mu}}_1^{(t)})$

(c) In the standard k-means algorithm, we assign $\widehat{\boldsymbol{\mu}}_1^{(t+1)} = \dfrac{\sum_{i=1}^{N} b_{i1}\boldsymbol{x}_i}{\sum_{i=1}^{N} b_{i1}}$.

$$\frac{\sum_{i=1}^{N} b_{i1}\boldsymbol{x}_i}{\sum_{i=1}^{N} b_{i1}} = \widehat{\boldsymbol{\mu}}_1^{(t)} + 2\eta\sum_{i=1}^{N} b_{i1}(\boldsymbol{x}_i - \widehat{\boldsymbol{\mu}}_1^{(t)}) \Rightarrow \eta = \frac{1}{2\sum_{i=1}^{N} b_{i1}}$$

Question 7:
One may be concerned that the randomness introduced in random forests may cause trouble, for instance, some features or samples may not be considered at all.

(a) Consider N training samples in a feature space of D dimensions. Consider building a random forest with T binary trees, each having exactly H internal nodes. Let F be the number of features randomly selected at each node. In order to simplify our calculations, we will let $F = 1$. For this setting, compute the probability that a certain feature (say, the first feature) is never considered for splitting.

(b) Now let us investigate the concern regarding the random selection of the samples. Suppose each tree employs N bootstrapped training samples. Compute the probability that a particular sample (say, the first sample) is never considered in any of the trees.

(c) Compute the values of the probabilities you obtained in the previous two parts for the case when there are $N = 2$ training samples, $D = 2$ dimensions, $T = 10$ trees of depth $H = 4$. What conclusions can you draw from your answer with regard to the concern mentioned at the beginning of the question?

(a) The probability that it is not considered for splitting in a particular node of a particular tree is $(1 - 1/D)$. The subsampling of $F = 1$ features at each node is independent of all others. There are a total of TH nodes and hence the final answer is $(1 - 1/D)^{TH}$.

(b) The probability that it is not considered in one of the trees is $(1 - 1/N)^N$. Since the choice for every tree is independent, the probability that it is not considered in any of the trees is $(1 - 1/N)^{NT}$.

(c) $(1/2)^{40}$ and $(1/2)^{20}$. It is quite unlikely that a feature or a sample will be missed.

Question 8:
It is suggested to run the k-means clustering algorithm multiple times with different initializations. Explain the reasoning behind this suggestion.

The k-means clustering algorithm is heavily affected by the initial configuration. That is why it is always a good idea to run the algorithm multiple times and pick the best solution.

Question 9:
Let a configuration of the k-means clustering algorithm correspond to the k way partition (on the set of instances to be clustered) generated by the clustering at the end of each iteration. Is it possible for the k-means algorithm to revisit a configuration? Justify your answer.

It is guaranteed that the objective function value of the k-means clustering algorithm is monotonically decreasing during successive iterations. That is why it is not possible to revisit a configuration once you move to another but a better configuration.

Question 10:
What are the distance and linkage functions used in hierarchical clustering algorithms? How do they differ in their roles?

Distance function: Distance measure between two instances $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$.

Linkage function: Distance measure between two groups $\mathcal{G}_A$ and $\mathcal{G}_B$.

Question 11:
What is an unstable learner? Why does bagging (i.e., bootstrap aggregating) rely on having an unstable learner as the base classifier?

A learning algorithm is unstable if small changes in the training set causes a large difference in the generated learner.

Bagging trains base-learners on similar data sets and, in order to generate diverse learners on these data sets, we need to have unstable learners.

Question 12:
What are the advantages of K-fold cross-validation and 5×2 cross-validation on each other?

K-fold cross-validation:

+ Training set size is large
− Overlap between training sets is large

5×2 cross-validation:

+ Overlap between training sets is small
− Training set size is small

Question 13:
For classification algorithms, there are different performance metrics such as classification accuracy, area under the ROC curve, precision, recall, etc. Explain when it is not a good idea to use the classification accuracy.

For heavily imbalanced data sets, it is not a good idea to use the classification accuracy as the performance metric since returning the majority class for all test samples will give a very high classification accuracy.

ENGR 421/DASC 521: Introduction to Machine Learning
Fall 2021 Midterm – Solution Key

Question 1:
"Introduction to Machine Learning" (ENGR 421/DASC 521) course is attended by students majoring in engineering and some students that do not major in engineering. 80% of the engineering students and 60% of the non-engineering students passed the course. 10% of the entire class are non-engineering students. What is the percentage of engineering students among those that actually passed the course?

Let S denote that a student passed the exam. Let E denote that a student is an engineering major, and N denote the otherwise.

$$\begin{aligned}\Pr(E) &= 0.9\\ \Pr(N) &= 0.1\\ \Pr(S|E) &= 0.8\\ \Pr(S|N) &= 0.6\end{aligned}$$

$$\begin{aligned}\Pr(E|S) =& \frac{\Pr(S|E)\Pr(E)}{\Pr(S)} = \frac{\Pr(S|E)\Pr(E)}{\Pr(S|E)\Pr(E)+\Pr(S|N)\Pr(N)}\\ =& \frac{0.8\times 0.9}{0.8\times 0.9+0.6\times 0.1}\\ =& \frac{72}{78} = \frac{12}{13} \approx 92.30\%\end{aligned}$$

Question 2:
Consider a data set in which each data point $\boldsymbol{x}_i \in \mathbb{R}^D$ is associated with a read-valued output $y_i \in \mathbb{R}$ and a weighting factor $r_i > 0$, so that the sum-of-squares error function becomes

$$E(\boldsymbol{w}) = \frac{1}{2}\sum_{i=1}^{N} r_i(y_i - \boldsymbol{w}^\top \boldsymbol{x}_i)^2.$$

Find an expression for the solution $\boldsymbol{w}^\star$ that minimizes this error function.

$$\frac{\partial E(\boldsymbol{w})}{\partial \boldsymbol{w}} = -\sum_{i=1}^{N} r_i(y_i - \boldsymbol{w}^\top \boldsymbol{x}_i)\boldsymbol{x}_i = 0$$

$$\begin{aligned}\sum_{i=1}^{N} r_i y_i \boldsymbol{x}_i &= \left(\sum_{i=1}^{N} r_i \boldsymbol{x}_i \boldsymbol{x}_i^\top\right)\boldsymbol{w}\\ \boldsymbol{w}^\star &= \left(\sum_{i=1}^{N} r_i \boldsymbol{x}_i \boldsymbol{x}_i^\top\right)^{-1}\left(\sum_{i=1}^{N} r_i y_i \boldsymbol{x}_i\right)\end{aligned}$$

Question 3:
Let us assume X is a random variable with the following probability distribution:

$$p(x) = 2\alpha x \exp(-\alpha x^2)$$

where x is a positive real number and α is a positive parameter. You are given a training data set consisting of N examples as $\mathcal{X} = \{x_i\}_{i=1}^N$.

(a) Describe a maximum likelihood approach to infer α an write down the log-likelihood objective for this problem.

(b) Find the maximum likelihood solution for α.

(a) By assuming the data points are independent from each other, we can write down the likelihood function as follows:

$$\text{likelihood} = \prod_{i=1}^{N} p(x_i) = \prod_{i=1}^{N} 2\alpha x_i \exp(-\alpha x_i^2)$$

$$\text{log-likelihood} = \sum_{i=1}^{N} \big(\log(2) + \log(\alpha) + \log(x_i) - \alpha x_i^2\big)$$

(b) To maximize log-likelihood, we need to minimize $\sum_{i=1}^{N}(\alpha x_i^2 - \log(\alpha))$ with respect to α.

$$\frac{\partial \sum_{i=1}^{N}(\alpha x_i^2 - \log(\alpha))}{\partial \alpha} = \sum_{i=1}^{N} \frac{\partial(\alpha x_i^2 - \log(\alpha))}{\partial \alpha}$$

$$= \sum_{i=1}^{N} \left(x_i^2 - \frac{1}{\alpha}\right)$$

$$\sum_{i=1}^{N} \left(x_i^2 - \frac{1}{\alpha}\right) = 0 \Rightarrow \frac{N}{\alpha^\star} = \sum_{i=1}^{N} x_i^2$$

$$\alpha^\star = \frac{N}{\sum_{i=1}^{N} x_i^2}$$

Question 4:
Let $p(x|y=1) \sim N(x; \mu_1, \sigma^2)$ and $p(x|y=2) \sim N(x; \mu_2, \sigma^2)$, that is, the variances are equal. For the linear discriminant

$$g(x) = \log \frac{P(y=1|x)}{P(y=2|x)} = wx + w_0,$$

write w and w_0 in terms of μ_1, μ_2, σ, and the prior probabilities, $P(y=1)$, $P(y=2)$.

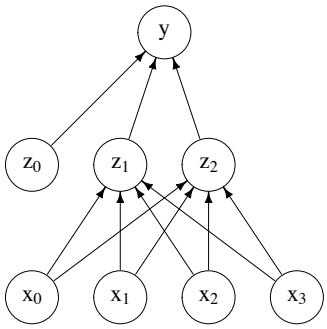
$$g(x) = \log \frac{P(y=1|x)}{P(y=2|x)} = \log \frac{p(x|y=1)P(y=1)}{p(x|y=2)P(y=2)}$$

$$
\begin{aligned}
&= \log \frac{\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu_1)^2}{2\sigma^2}\right) P(y=1)}{\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu_2)^2}{2\sigma^2}\right) P(y=2)} \\
&= -\frac{(x-\mu_1)^2}{2\sigma^2} + \frac{(x-\mu_2)^2}{2\sigma^2} + \log \frac{P(y=1)}{P(y=2)} \\
&= \underbrace{\frac{\mu_1-\mu_2}{\sigma^2}}_{w} x + \underbrace{\frac{\mu_2^2-\mu_1^2}{2\sigma^2} + \log \frac{P(y=1)}{P(y=2)}}_{w_0}
\end{aligned}
$$

Question 5:
You are asked to design a multilayer perceptron which receives three binary-valued (i.e., 0 or 1) inputs x_1, x_2, and x_3, and outputs 1 if exactly two of the inputs are 1, and outputs 0 otherwise. All of the units use a hard threshold activation function:

$$
s(a) = \begin{cases} 1 & \text{if } a \geq 0, \\ 0 & \text{if } a < 0. \end{cases}
$$

Specify weights which correctly implement this function. You do not need to explain your solution. (Hint: one of the hidden units should activate if two or more inputs are on, and the other should activate if all of the inputs are on.)



$$
\mathbf{W} = \begin{bmatrix} w_{10} & w_{11} & w_{12} & w_{13} \\ w_{20} & w_{21} & w_{22} & w_{23} \end{bmatrix} = \begin{bmatrix} -1.5 & +1 & +1 & +1 \\ -2.5 & +1 & +1 & +1 \end{bmatrix}
$$
$$
\boldsymbol{v} = \begin{bmatrix} v_0 & v_1 & v_2 \end{bmatrix} = \begin{bmatrix} -0.5 & +1 & -1 \end{bmatrix}
$$

Question 6:
Does it make sense to initialize all weights in a multilayer perceptron to zero? Justify your answer.

No. If all weights are equal, their gradients will be the same as well, and the gradients will possibly vanish. Hence all neurons will learn the same feature.

Question 7:
Assume we have a set of data from patients who have visited a hospital during the year. A set of features (e.g., temperature, height, etc.) have been also extracted for each patient. Our goal is to decide whether a new visiting patient has any of diabetes, heart disease, or Alzheimer (a patient can have one or more of these diseases). We have decided to use a neural network to solve this problem. We have two choices: either to train a separate neural network for each of the diseases or to train a single neural network with one output neuron for each disease, but with a shared hidden layer. Which method do you prefer? Justify your answer.

Neural network with a shared hidden layer can capture dependencies between diseases. It can be shown that in some cases, when there is a dependency between the output nodes, having a shared node in the hidden layer can improve the accuracy.

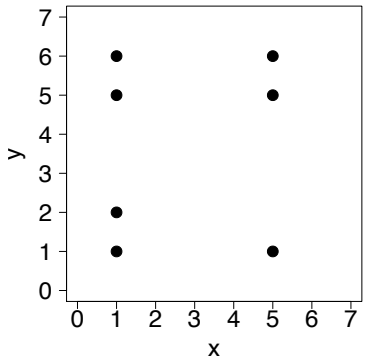
If there is no dependency between diseases (output neurons), then we would prefer to have a separate neural network for each disease.

Question 8:
List two drawbacks of the naive density estimator. What is the advantage of the Parzen windows density estimator compared to the naive density estimator?

Not continuous, curse of dimensionality, zero density if no training points in the bin, etc.

The Parzen windows density estimator gives smoother probability density functions than the naive density estimator.

Question 9:
The following figure shows a data set with one-real valued input x and one real-valued output y. There are seven training samples.



Suppose you are training a kernel smoother using some unspecified kernel function. The only thing you know about the kernel function is that it is a monotonically decreasing function of distance that decays to zero at a distance of 3 units (and is strictly greater than zero at a distance of less than 3 units).

(a) What is the predicted value of y when $x = 1$? Justify your answer.

(b) What is the predicted value of y when $x = 3$? Justify your answer.

(c) What is the predicted value of y when $x = 4$? Justify your answer.

(a) $\widehat{y} = (1+2+5+6)/4$ since the distance to data points at $x = 5$ is 4 (they have no effect) and the distance to data points at $x = 1$ is 0 (they have equal effects).

(b) $\widehat{y} = (1+2+5+6+1+5+6)/7$ since the distance to all data points is 2 (they have equal effects).

(c) $\widehat{y} = (1+5+6)/3$ since the distance to data points at $x = 1$ is 3 (they have no effect) and the distance to data points at $x = 5$ is 1 (they have equal effects).

Question 10:
The naive density estimator $\widehat{p}(x)$ for data point $x \in \mathbb{R}$ is given by

$$\widehat{p}(x) = \frac{1}{Nh} \sum_{i=1}^{N} w\left(\frac{x - x_i}{h}\right)$$

$$w(u) = \begin{cases} 1 & \text{if } |u| \leq 1/2 \\ 0 & \text{otherwise} \end{cases}$$

where the training data set consists of training samples $x_1, x_2, \ldots, x_N$.

(a) Discuss how the value of h affects the naive density estimator. How should the value of h be varied as N changes?

(b) Show that the naive density estimator $\widehat{p}(x)$ is indeed a valid probability density function.

(a) small $h \Rightarrow$ overfitting, large $h \Rightarrow$ underfitting

small $N \Rightarrow$ we can increase h, large $N \Rightarrow$ we can decrease h

(b) $\widehat{p}(x) \geq 0$ is trivially satisfied.

$$\begin{aligned}
\int_{x\in\mathbb{R}} \widehat{p}(x)\mathrm{d}x &= \int_{x\in\mathbb{R}} \frac{1}{Nh}\sum_{i=1}^{N} w\left(\frac{x-x_i}{h}\right)\mathrm{d}x \\
&= \frac{1}{Nh}\sum_{i=1}^{N}\int_{x\in\mathbb{R}} w\left(\frac{x-x_i}{h}\right)\mathrm{d}x \\
&= \frac{1}{Nh}\sum_{i=1}^{N}\left[\int_{x=-\infty}^{x_i-h/2} 0\mathrm{d}x + \int_{x=x_i-h/2}^{x_i+h/2} 1\mathrm{d}x + \int_{x=x_i+h/2}^{+\infty} 0\mathrm{d}x\right] \\
&= \frac{1}{Nh}\sum_{i=1}^{N}[0+h+0] = \frac{1}{Nh}Nh = 1
\end{aligned}$$

Question 11:
Instead of entropy in decision trees, one can use the Gini index or the misclassification error. You have a data set with 400 positive examples (i.e., 400^+) and 400 negative examples (i.e., 400^-). Suppose that you have two possible splits for a decision node m. The first splitting choice results in L_m: $(300^+, 100^-)$ and R_m: $(100^+, 300^-)$. The second splitting choice results in L_m: $(200^+, 400^-)$ and R_m: $(200^+, 0^-)$.

(a) Calculate the impurity after these two choices of splitting using the Gini index and the misclassification error?

(b) Which of these two choices of splitting would be preferred and why?

(a) Gini index $= 2p(1-p)$ Misclassification error $= \min(p, 1-p)$

$$\begin{aligned}
\text{Gini}(1^{st}\text{ split}) &= \frac{400}{800}\left(2\frac{300}{400}\frac{100}{400}\right) + \frac{400}{800}\left(2\frac{100}{400}\frac{300}{400}\right) = \frac{3}{8} \\
\text{Gini}(2^{nd}\text{ split}) &= \frac{600}{800}\left(2\frac{200}{600}\frac{400}{600}\right) + \frac{200}{800}\left(2\frac{200}{200}\frac{0}{200}\right) = \frac{1}{3} \\
\text{Error}(1^{st}\text{ split}) &= \frac{400}{800}\min\left(\frac{300}{400}, \frac{100}{400}\right) + \frac{400}{800}\min\left(\frac{100}{400}, \frac{300}{400}\right) = \frac{1}{4} \\
\text{Error}(2^{nd}\text{ split}) &= \frac{600}{800}\min\left(\frac{200}{600}, \frac{400}{600}\right) + \frac{200}{800}\min\left(\frac{200}{200}, \frac{0}{200}\right) = \frac{1}{4}
\end{aligned}$$

(b) Gini index prefers the second split, whereas there is no difference in terms of misclassification error.

Question 12:
Explain the difference between prepruning and postpruning strategies for decision trees. Why does postpruning work better than prepruning in practice?

Stopping the tree construction early on before it is full is called prepruning the tree. In postpruning, we try to find and prune unnecessary subtrees. We grow the tree full until all leaves are pure and we have no training error. We then find subtrees that cause overfitting and we prune them.

Postpruning generally leads to more accurate trees since, from the initial labeled set, we set aside a pruning set, unused during training, and postpruning is performed by looking at the performance on this set.

ENGR 421/DASC 521: Introduction to Machine Learning
Spring 2021 Final – Solution Key

Question 1:
What are the differences between parametric methods and non-parametric methods? Name some of the parametric and non-parametric machine learning algorithms.

- **Parametric**: Number of parameters are not affected by the training data set size, stronger assumptions about the data
  Examples: linear/logistic regression, linear discriminant analysis, perceptron, neural networks
- **Non-parametric**: Number of parameters grows with the size of the training set, fewer assumptions about the data
  Examples: k-nearest neighbours, decision trees, support vector machines

Question 2:
For a split used for a numeric input in a decision tree, instead of a binary split, how can we increase the number of children to three? What are the advantages and the disadvantages of such a split over a binary split?

One can use a ternary split with two thresholds and three branches as

$$x_j < w_{ma}, \quad w_{ma} \leq x_j < w_{mb}, \quad x_j \geq w_{mb}.$$

- One ternary node splits an input into three, whereas this requires two successive binary nodes.
- The complexity of finding the best pair grows larger and each node stores two thresholds instead of one and has three branches instead of two.

Question 3:
In regression problems, we calculate the error as the sum of squared differences between the true and predicted values. This particular choice is the most frequently used error function, but it is not robust to outliers since it increases rapidly with the increasing size of the difference.

(a) What would be a better error function to implement robust regression?

(b) Why do we usually prefer working with the sum of squared differences?

(a) One alternative is to use the absolute value (or ϵ-insensitive loss) as an error function instead of squaring the differences.

(b) This achieves robustness, but is hard to work with in practice because the absolute value function is not differentiable.

Question 4:
Suppose that you are given the source code of a decision tree algorithm for classification. Describe how you can modify the source code to obtain a decision tree algorithm for regression.

First, we should find an impurity measure suitable for regression problems such as squared error. We should then find a suitable stopping condition for determining leaf nodes.

Question 5:
Explain the effect of the dimensionality of the reduced space on the reconstruction error in principal component analysis (PCA).

As the dimensionality of the reduced space increases, the reconstruction error decreases. However, this decrease is very fast at first and slows later on.

Question 6:
What is the difference between principal component analysis (PCA) and linear discriminant analysis (LDA)?

Principal Component Analysis: It is an unsupervised learning method that aims to find the directions that maximize the variance of the data.

Linear Discriminant Analysis: It is a supervised learning method that aims to find the directions that best separate the classes.

Question 7:
Clustering is an unsupervised and iterative algorithm that has two steps:

(1) Estimate the labels,

(2) Estimate the parameters.

k-means and Expectation–Maximization (EM) are special cases of clustering algorithm. What are the labels and the parameters for these algorithms in (1) and (2), respectively.

For k-means, labels are cluster memberships (it is a binary variable), and parameters are cluster means (i.e., sample mean of each cluster).

For EM, labels are cluster memberships (it is a random variable), and parameters are cluster means and covariances (i.e., sample mean and covariances of each cluster).

Question 8:
Assume that we have two classes that are exponentially distributed with rates λ_1 and λ_2, and prior probabilities $P(y = 1)$ and $P(y = 2)$. The probability density function of exponential distribution with rate λ is $p(x) = \lambda e^{-\lambda x}$. Determine the parameters of the following discriminant function.

$$g(x) = \log \frac{P(y = 1|x)}{P(y = 2|x)} = w_1 x + w_0$$

$$
\begin{aligned}
g(x) &= \log\left[\frac{P(y=1|x)}{P(y=2|x)}\right] \\
&= \log\left[\frac{p(x|y=1)P(y=1)}{p(x|y=2)P(y=2)}\right] \\
&= \log\left[\frac{\lambda_1 e^{-\lambda_1 x} P(y=1)}{\lambda_2 e^{-\lambda_2 x} P(y=2)}\right] \\
&= \log\left[e^{(\lambda_2-\lambda_1)x}\right] + \log\left[\frac{\lambda_1 P(y=1)}{\lambda_2 P(y=2)}\right] \\
&= \underbrace{(\lambda_2-\lambda_1)}_{w_1} x + \underbrace{\log\left[\frac{\lambda_1 P(y=1)}{\lambda_2 P(y=2)}\right]}_{w_0}
\end{aligned}
$$

$$
\begin{aligned}
w_1 &= \lambda_2 - \lambda_1 \\
w_0 &= \log\left[\frac{\lambda_1 P(y=1)}{\lambda_2 P(y=2)}\right]
\end{aligned}
$$

Question 9:
Let $\mathcal{X} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^N$ be the training set that contains N data points for a binary classification problem. The primal optimization problem for support vector machine can be written as follows:

$$
\begin{aligned}
\text{minimize} \quad & \frac{1}{2}\|\boldsymbol{w}\|_2^2 + C\sum_{i=1}^N \gamma_i \xi_i \\
\text{with respect to} \quad & \boldsymbol{w},\ \boldsymbol{\xi},\ b \\
\text{subject to} \quad & y_i\big(\boldsymbol{w}^\top \boldsymbol{x}_i + b\big) \geq 1 - \xi_i \quad \forall i \\
& \xi_i \geq 0 \quad \forall i
\end{aligned}
$$

where $\boldsymbol{\xi}$ is the vector of slack variables, C is the regularization parameter, and γ_i is a given coefficient for balancing the effect of errors for different classes based on the class populations. Derive the dual optimization problem.

$$
\mathcal{L}_P = \frac{1}{2}\|\boldsymbol{w}\|_2^2 + C\sum_{i=1}^N \gamma_i \xi_i - \sum_{i=1}^N \alpha_i\big(y_i\big(\boldsymbol{w}^\top \boldsymbol{x}_i + b\big) - 1 + \xi_i\big) - \sum_{i=1}^N \beta_i \xi_i
$$

$$
\begin{aligned}
\frac{\partial \mathcal{L}_P}{\partial \boldsymbol{w}} &= \boldsymbol{w} - \sum_{i=1}^N \alpha_i y_i \boldsymbol{x}_i = 0 \quad &\Rightarrow \quad \boldsymbol{w} &= \sum_{i=1}^N \alpha_i y_i \boldsymbol{x}_i \\
\frac{\partial \mathcal{L}_P}{\partial b} &= -\sum_{i=1}^N \alpha_i y_i = 0 \quad &\Rightarrow \quad \sum_{i=1}^N \alpha_i y_i &= 0 \\
\frac{\partial \mathcal{L}_P}{\partial \xi_i} &= C\gamma_i - \alpha_i - \beta_i = 0 \quad \forall i \quad &\Rightarrow \quad 0 \leq \alpha_i &\leq C\gamma_i \quad \forall i
\end{aligned}
$$

$$
\text{maximize} \quad \sum_{i=1}^N \alpha_i - \frac{1}{2}\sum_{i=1}^N\sum_{j=1}^N \alpha_i \alpha_j y_i y_j \boldsymbol{x}_i^\top \boldsymbol{x}_j
$$

$$\begin{aligned} \text{with respect to} \quad & \boldsymbol{\alpha} \\ \text{subject to} \quad & \sum_{i=1}^{N} \alpha_i y_i = 0 \\ & 0 \leq \alpha_i \leq C\gamma_i \quad \forall i \end{aligned}$$

Question 10:
Assume that we have two classes from two multivariate Gaussian distributions with different mean vectors and equal covariance matrices; that is, we have $p(\boldsymbol{x}|y=1) \sim N(\boldsymbol{\mu}_1, \boldsymbol{\Sigma})$ and $p(\boldsymbol{x}|y=2) \sim N(\boldsymbol{\mu}_2, \boldsymbol{\Sigma})$. Assume also that the prior probabilities are equal; that is, we have $P(y=1) = P(y=2)$. Derive the position of the intersection of the two posterior probabilities.

$$\begin{aligned} P(y=1|\boldsymbol{x}) &= P(y=2|\boldsymbol{x}) \\ p(\boldsymbol{x}|y=1)\underbrace{P(y=1)}_{1/2} &= p(\boldsymbol{x}|y=2)\underbrace{P(y=2)}_{1/2} \\ \frac{1}{\sqrt{(2\pi)^D|\boldsymbol{\Sigma}|}} \exp\left[-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_1)^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_1)\right] &= \frac{1}{\sqrt{(2\pi)^D|\boldsymbol{\Sigma}|}} \exp\left[-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_2)^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_2)\right] \\ (\boldsymbol{x}-\boldsymbol{\mu}_1)^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_1) &= (\boldsymbol{x}-\boldsymbol{\mu}_2)^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_2) \\ (\boldsymbol{x}-\boldsymbol{\mu}_1) &= -(\boldsymbol{x}-\boldsymbol{\mu}_2) \\ \boldsymbol{x}^\star &= \frac{\boldsymbol{\mu}_1+\boldsymbol{\mu}_2}{2} \end{aligned}$$

Question 11:
Let $\mathcal{X} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^N$ be the training set that contains N multivariate data points for a linear regression problem. Let $\mathcal{L}(\boldsymbol{w}, b)$ be the regularized loss function:

$$\mathcal{L}(\boldsymbol{w}, b) = \frac{1}{N}\sum_{i=1}^{N} l(y_i - \boldsymbol{w}^\top \boldsymbol{x}_i - b) + \frac{\lambda}{2}\|\boldsymbol{w}\|_2^2$$

where λ is a regularization constant, and $l(\cdot)$ is defined as

$$l(\xi) = \begin{cases} \frac{1}{2}\xi^2 & \text{if } |\xi| < 1 \\ |\xi| - \frac{1}{2} & \text{otherwise.} \end{cases}$$

Calculate the gradient of the regularized loss function with respect to $\boldsymbol{w}$.

$$\frac{\partial l(\xi)}{\partial \xi} = \begin{cases} \xi & \text{if } |\xi| < 1 \\ \text{sign}(\xi) & \text{otherwise} \end{cases}$$

$$\frac{\partial \mathcal{L}(\boldsymbol{w}, b)}{\partial \boldsymbol{w}} = \frac{1}{N}\sum_{i=1}^{N} \frac{\partial l(y_i - \boldsymbol{w}^\top \boldsymbol{x}_i - b)}{\partial \boldsymbol{w}} + \lambda \boldsymbol{w}$$

$$\frac{\partial l(y_i - \boldsymbol{w}^\top \boldsymbol{x} - b)}{\partial \boldsymbol{w}} = \begin{cases} (y_i - \boldsymbol{w}^\top \boldsymbol{x}_i - b)(-\boldsymbol{x}_i) & \text{if } |y_i - \boldsymbol{w}^\top \boldsymbol{x}_i - b| < 1 \\ \text{sign}(y_i - \boldsymbol{w}^\top \boldsymbol{x}_i - b)(-\boldsymbol{x}_i) & \text{otherwise} \end{cases}$$

Question 12:
Consider the following training data set for a binary classification problem:

positively labeled data points: $(3, 1)$, $(3, -1)$, $(6, 1)$, $(6, -1)$

negatively labeled data points: $(1, 0)$, $(0, 1)$, $(0, -1)$, $(-1, 0)$

Write the primal and dual support vector machine problems corresponding to this problem instance. Give the optimal solutions to the primal and dual optimization problems. Identify the support vectors. Write the separating hyperplane in terms of the support vectors.

$$\begin{aligned} \text{minimize} \quad & \frac{1}{2}(w_1^2 + w_2^2) \\ \text{with respect to} \quad & w_1,\ w_2,\ b \\ \text{subject to} \quad & +1(+3w_1 + 1w_2 + b) \geq 1 \\ & +1(+3w_1 - 1w_2 + b) \geq 1 \\ & +1(+6w_1 + 1w_2 + b) \geq 1 \\ & +1(+6w_1 - 1w_2 + b) \geq 1 \\ & -1(+1w_1 + 0w_2 + b) \geq 1 \\ & -1(+0w_1 + 1w_2 + b) \geq 1 \\ & -1(+0w_1 - 1w_2 + b) \geq 1 \\ & -1(-1w_1 + 0w_2 + b) \geq 1 \end{aligned}$$

$w_1^\star = 1,\ w_2^\star = 0,\ b^\star = -2$

$$\begin{aligned} \text{maximize} \quad & \sum_{i=1}^{8} \alpha_i - \frac{1}{2}\begin{bmatrix}\alpha_1 & \alpha_2 & \dots & \alpha_8\end{bmatrix} \begin{bmatrix} y_1y_1\boldsymbol{x}_1^\top\boldsymbol{x}_1 & y_1y_2\boldsymbol{x}_1^\top\boldsymbol{x}_2 & \dots & y_1y_8\boldsymbol{x}_1^\top\boldsymbol{x}_8 \\ y_2y_1\boldsymbol{x}_2^\top\boldsymbol{x}_1 & y_2y_2\boldsymbol{x}_2^\top\boldsymbol{x}_2 & \dots & y_2y_8\boldsymbol{x}_2^\top\boldsymbol{x}_8 \\ \vdots & \vdots & \ddots & \vdots \\ y_8y_1\boldsymbol{x}_8^\top\boldsymbol{x}_1 & y_8y_2\boldsymbol{x}_8^\top\boldsymbol{x}_2 & \dots & y_8y_8\boldsymbol{x}_8^\top\boldsymbol{x}_8 \end{bmatrix} \begin{bmatrix}\alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_8\end{bmatrix} \\ \text{with respect to} \quad & \alpha_1,\ \alpha_2,\ \alpha_3,\ \alpha_4,\ \alpha_5,\ \alpha_6,\ \alpha_7,\ \alpha_8 \\ \text{subject to} \quad & +\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 - \alpha_5 - \alpha_6 - \alpha_7 - \alpha_8 = 0 \\ & \alpha_i \geq 0 \quad i = 1, 2, \dots, 8 \end{aligned}$$

$\alpha_1^\star = 1/4,\ \alpha_2^\star = 1/4,\ \alpha_3^\star = 0,\ \alpha_4^\star = 0,\ \alpha_5^\star = 1/2,\ \alpha_6^\star = 0,\ \alpha_7^\star = 0,\ \alpha_8^\star = 0$

Question 13:
Let $\mathcal{X} = \{(x_i, y_i)\}_{i=1}^N$ be the training set that contains N univariate data points for a regression problem, which is described by the following model:

$$y_i = ax_i + \epsilon_i$$

where $\epsilon_i \sim \mathcal{N}(\epsilon_i; 0, \sigma^2)$, and each data point is independent of the others.

(a) Write the likelihood function $p(y_1, y_2, \dots, y_N | x_1, x_2, \dots, x_N, a, \sigma^2)$.

(b) Compute the maximum likelihood estimate of a.

$$\epsilon_i \sim \mathcal{N}(\epsilon_i; 0, \sigma^2) \Rightarrow y_i \sim \mathcal{N}(y_i; ax_i, \sigma^2)$$

(a)

$$\begin{aligned} p(y_1, y_2, \ldots, y_N | x_1, x_2, \ldots, x_N, a, \sigma^2) &= \prod_{i=1}^{N} p(y_i | x_i, a, \sigma^2) \\ &= \prod_{i=1}^{N} \left[\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - ax_i)^2}{2\sigma^2} \right) \right] \end{aligned}$$

(b)

$$\begin{aligned} \text{log-likelihood} &= \sum_{i=1}^{N} \log \left[\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - ax_i)^2}{2\sigma^2} \right) \right] \\ &=^{+} \sum_{i=1}^{N} -\frac{(y_i - ax_i)^2}{2\sigma^2} \end{aligned}$$

$$\frac{\partial \text{log-likelihood}}{\partial a} = \sum_{i=1}^{N} -\frac{2(y_i - ax_i)(-x_i)}{2\sigma^2} = 0 \Rightarrow a^\star = \frac{\sum_{i=1}^{N} x_i y_i}{\sum_{i=1}^{N} x_i^2}$$

ENGR 421/DASC 521: Introduction to Machine Learning
Spring 2021 Midterm – Solution Key

Question 1:
Explain why it is not a good idea to use standard regression algorithms on classification problems. What makes the logistic regression algorithm more suitable for classification problems?

Regression algorithms may produce arbitrary outputs (i.e., smaller than 0 or greater than 1). However, logistic regression algorithm is guaranteed to produce probability values between 0 and 1.

Question 2:
Suppose that you are given the source code of a binary classification algorithm. Describe how you can use the same source code to obtain a multiclass classification algorithm.

We can use the source code of a binary classification algorithm to obtain a multiclass classification algorithm either by one-versus-all or one-versus-other approach.

Question 3:
The softmax function is defined as

$$\widehat{y}_c = \frac{\exp(a_c)}{\sum_{b=1}^{K} \exp(a_b)}.$$

In softmax implementations, we usually use the following formula:

$$\widehat{y}_c = \frac{\exp(a_c - m)}{\sum_{b=1}^{K} \exp(a_b - m)},$$

where $m = \max(a_1, \ldots, a_K)$. What is the purpose of this modification?

We use this modification to avoid overflow problems when calculating the softmax function.

Question 4:
It is always a good idea to initialize the weights in linear discrimination with random values close to 0. They are generally drawn uniformly from the interval $[-0.01, 0.01]$. What is the reason for such an initialization strategy?

If the initial weights are large in magnitude, the weighted sum may also be large and may saturate the sigmoid. If the initial weights are close to 0, the weighted sum will stay close to 0 where the derivative is nonzero and an update can take place. If the weighted sum is large in magnitude, the derivative of the sigmoid will be almost 0 and weights will not be updated.

Question 5:
In gradient descent, we use a constant learning rate η that is the same for all weights and for all epochs. How can we make it adaptive?

In gradient descent, the learning factor η determines the magnitude of change to be made in the parameter. A very early heuristic was to make it adaptive for faster convergence, where it is kept large when learning takes place and is decreased when learning slows down. Thus we increase η by a constant amount (or keep it unchanged) if the error on the training set decreases and decrease it geometrically if it increases.
In a deep network with many hidden layers and units, the contributions of weights to the error are not the same, and one can adapt the learning factor separately for each weight, depending on the convergence along that direction. The idea in such methods is to accumulate the past error gradient magnitudes for each weight and then make the learning factor inversely proportional to that. In directions where the gradient is small, the learning factor is larger to compensate, and it can be smaller along directions where the gradient is already large.

Question 6:
Consider a multilayer perceptron architecture with one hidden layer where there are also direct weights from the inputs directly to the output units. Explain when such a structure would be helpful and how it can be trained.

It can be trained as usual by backpropagation. The direct weights from the input to the outputs is a perceptron and the rest is a multilayer perceptron, each being trained together as usual. The perceptron that contains the direct weights from the inputs to the outputs is a linear model and can be thought of as implementing the linear component of whatever we are trying to learn; the multilayer perceptron then learns to correct this default linear model.

Question 7:
What are the advantages of using the Mahalanobis distance over the Euclidean distance?

- Mahalanobis distance: $\sqrt{(\boldsymbol{x}-\widehat{\boldsymbol{\mu}})^\top \widehat{\boldsymbol{\Sigma}}^{-1}(\boldsymbol{x}-\widehat{\boldsymbol{\mu}})}$
- Euclidean distance: $\sqrt{(\boldsymbol{x}-\widehat{\boldsymbol{\mu}})^\top (\boldsymbol{x}-\widehat{\boldsymbol{\mu}})}$

The Mahalanobis distance normalizes features based on variance.

- This makes all independent attributes equally important.
- It alleviates problem caused by using different scales.
- It downplays the contribution of correlated attributes in distance computations.

Question 8:
In generating a univariate tree, a discrete attribute with K possible values can be represented by K 0/1 dummy variable and then treated as K separate numeric attributes. What are the advantages and disadvantages of this approach?

A discrete variable:

- Split decisions are complex.
- Tree size is small.

K binary variables:

- Split decisions become easier.
- Tree may grow large.

Question 9:
Consider a data set in which each data point $\boldsymbol{x}_i \in \mathbb{R}^D$ is associated with a real-valued output $y_i \in \mathbb{R}$. Instead of using the original sum-of-squares error function, we modify the error function by adding a regularization term:

$$E(\boldsymbol{w}) = \frac{1}{2}\sum_{i=1}^{N}(y_i - \boldsymbol{w}^\top \boldsymbol{x}_i)^2 + \frac{1}{2}\lambda \boldsymbol{w}^\top \boldsymbol{w},$$

where $\lambda > 0$ is the regularization coefficient.

(a) Find an expression for the solution $\boldsymbol{w}^\star$ that minimizes this error function.

(b) Describe a method to choose the best λ parameter from a set of candidate values.

(a) We should take the derivative of the error function and set it equal to zero.

$$\frac{\partial E(\boldsymbol{w})}{\partial \boldsymbol{w}} = -\sum_{i=1}^{N}(y_i - \boldsymbol{w}^\top \boldsymbol{x}_i)\boldsymbol{x}_i + \lambda \boldsymbol{w} = 0$$

$$\sum_{i=1}^{N} y_i \boldsymbol{x}_i = \left(\sum_{i=1}^{N} \boldsymbol{x}_i \boldsymbol{x}_i^\top + \lambda \mathbf{I}\right)\boldsymbol{w}$$

$$\boldsymbol{w}^\star = \left(\sum_{i=1}^{N} \boldsymbol{x}_i \boldsymbol{x}_i^\top + \lambda \mathbf{I}\right)^{-1}\left(\sum_{i=1}^{N} y_i \boldsymbol{x}_i\right)$$

(b) We should check the performance on the validation set(s) to pick the best λ parameter.

Question 10:
You are given a univariate binary classification data set. The first class has the distribution $p(x|y=1) \sim \mathcal{N}(x; \mu_1, \sigma^2)$, and the second class has the distribution $p(x|y=2) \sim \mathcal{N}(x; \mu_2, \sigma^2)$. The prior probabilities of each class are $P(y=1) = P(y=2) = 1/2$. Show that the posterior probability $P(y=1|x)$ is of the form

$$P(y=1|x) = \frac{1}{1+\exp(-wx - w_0)}$$

and determine w and w_0 in terms of μ_1, μ_2, and σ^2.

$$\log\left[\frac{P(y=1|x)}{1-P(y=1|x)}\right] = \log\left[\frac{p(x|y=1)P(y=1)/p(x)}{p(x|y=2)P(y=2)/p(x)}\right]$$

$$= \log\left[\frac{\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(x-\mu_1)^2}{2\sigma^2}\right)}{\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(x-\mu_2)^2}{2\sigma^2}\right)}\right]$$

$$
\begin{aligned}
&= \frac{(\mu_1 - \mu_2)}{\sigma^2} x + \frac{(\mu_2^2 - \mu_1^2)}{2\sigma^2} \\
P(y = 1|x) &= \frac{1}{1 + \exp\left[-\frac{(\mu_1 - \mu_2)}{\sigma^2} x - \frac{(\mu_2^2 - \mu_1^2)}{2\sigma^2}\right]} \\
w &= \frac{(\mu_1 - \mu_2)}{\sigma^2} \\
w_0 &= \frac{(\mu_2^2 - \mu_1^2)}{2\sigma^2}
\end{aligned}
$$

Question 11:
You are asked to design a text classifier to classify documents as either "sports" or "politics". You decide to represent each document as a vector of attributes describing the presence or absence of words.

$$\boldsymbol{x} = (\text{"goal"}, \text{"defence"}, \text{"office"}, \text{"strategy"})$$

The training data set constructed from eight documents is given below.

$$
\begin{aligned}
\boldsymbol{x}_1 &= (1,0,0,0) & y_1 &= \text{"sports"} \\
\boldsymbol{x}_2 &= (1,1,1,0) & y_2 &= \text{"sports"} \\
\boldsymbol{x}_3 &= (1,1,0,1) & y_3 &= \text{"sports"} \\
\boldsymbol{x}_4 &= (1,1,1,1) & y_4 &= \text{"politics"} \\
\boldsymbol{x}_5 &= (0,1,1,1) & y_5 &= \text{"politics"} \\
\boldsymbol{x}_6 &= (0,1,1,0) & y_6 &= \text{"politics"} \\
\boldsymbol{x}_7 &= (0,0,0,1) & y_7 &= \text{"politics"} \\
\boldsymbol{x}_8 &= (0,1,0,1) & y_8 &= \text{"politics"}
\end{aligned}
$$

Using a naive Bayes classifier trained with a maximum likelihood approach, what is the probability that a new document $\boldsymbol{x} = (1,0,1,0)$ is about "politics"?

$$
\begin{aligned}
\widehat{P}(\text{sports}) &= 3/8 & \widehat{P}(\text{politics}) &= 5/8 \\
\widehat{P}(\texttt{1...}|\text{sports}) &= 3/3 & \widehat{P}(\texttt{1...}|\text{politics}) &= 1/5 \\
\widehat{P}(\texttt{.1..}|\text{sports}) &= 2/3 & \widehat{P}(\texttt{.1..}|\text{politics}) &= 4/5 \\
\widehat{P}(\texttt{..1.}|\text{sports}) &= 1/3 & \widehat{P}(\texttt{..1.}|\text{politics}) &= 3/5 \\
\widehat{P}(\texttt{...1}|\text{sports}) &= 1/3 & \widehat{P}(\texttt{...1}|\text{politics}) &= 4/5 \\
\widehat{P}(\texttt{0...}|\text{sports}) &= 0/3 & \widehat{P}(\texttt{0...}|\text{politics}) &= 4/5 \\
\widehat{P}(\texttt{.0..}|\text{sports}) &= 1/3 & \widehat{P}(\texttt{.0..}|\text{politics}) &= 1/5 \\
\widehat{P}(\texttt{..0.}|\text{sports}) &= 2/3 & \widehat{P}(\texttt{..0.}|\text{politics}) &= 2/5 \\
\widehat{P}(\texttt{...0}|\text{sports}) &= 2/3 & \widehat{P}(\texttt{...0}|\text{politics}) &= 1/5
\end{aligned}
$$

$$
\begin{aligned}
\widehat{P}(\text{sports}|\texttt{1010}) &\propto (3/3)(1/3)(1/3)(2/3)(3/8) \\
\widehat{P}(\text{politics}|\texttt{1010}) &\propto (1/5)(1/5)(3/5)(1/5)(5/8) \\
\widehat{P}(\text{politics}|\texttt{1010}) &\approx 0.0975
\end{aligned}
$$

Question 12:
Show that the derivative of the softmax

$$\widehat{y}_c = \frac{\exp(a_c)}{\sum_{b=1}^{K} \exp(a_b)}$$

is $\partial \widehat{y}_c / \partial a_d = \widehat{y}_c(\delta_{cd} - \widehat{y}_d)$, where δ_{cd} is 1 if $c = d$ and 0 otherwise.

$$\frac{\partial \widehat{y}_c}{\partial a_d} = \frac{\frac{\partial \exp(a_c)}{\partial a_d}\left(\sum_{b=1}^{K} \exp(a_b)\right) - \exp(a_c)\frac{\partial\left(\sum_{b=1}^{K} \exp(a_b)\right)}{\partial a_d}}{\left(\sum_{b=1}^{K} \exp(a_b)\right)^2}$$

$$= \frac{\exp(a_c)\delta_{cd}}{\sum_{b=1}^{K} \exp(a_b)} - \frac{\exp(a_c)}{\sum_{b=1}^{K} \exp(a_b)} \frac{\exp(a_d)}{\sum_{b=1}^{K} \exp(a_b)} = \widehat{y}_c\delta_{cd} - \widehat{y}_c\widehat{y}_d = \widehat{y}_c(\delta_{cd} - \widehat{y}_d)$$

Question 13:
The kernel density estimator $\widehat{p}(\boldsymbol{x})$ for a multivariate data point $\boldsymbol{x} \in \mathbb{R}^D$ is written as follows:

$$\widehat{p}(\boldsymbol{x}) = \frac{1}{Nh^D} \sum_{i=1}^{N} K\left(\frac{\boldsymbol{x} - \boldsymbol{x}_i}{h}\right)$$

$$K(\boldsymbol{u}) = \frac{1}{\sqrt{(2\pi)^D}} \exp\left(-\frac{\boldsymbol{u}^\top \boldsymbol{u}}{2}\right)$$

where the training data set consists of training samples $\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N$.

(a) Discuss how the value of h affects the kernel density estimator. How should the value of h be varied as N changes?

(b) Show that the kernel density estimator $\widehat{p}(\boldsymbol{x})$ is indeed a valid probability density function.

(a) small $h \Rightarrow$ overfitting, large $h \Rightarrow$ underfitting

small $N \Rightarrow$ we can increase h, large $N \Rightarrow$ we can decrease h

(b) $\widehat{p}(\boldsymbol{x}) \geq 0$ is trivially satisfied.

$$\int_{x_1 \in \mathbb{R}} \cdots \int_{x_D \in \mathbb{R}} \widehat{p}(\boldsymbol{x}) \mathrm{d}x_1 \ldots \mathrm{d}x_D = \int_{x_1 \in \mathbb{R}} \cdots \int_{x_D \in \mathbb{R}} \frac{1}{Nh^D} \sum_{i=1}^{N} K\left(\frac{\boldsymbol{x} - \boldsymbol{x}_i}{h}\right) \mathrm{d}x_1 \ldots \mathrm{d}x_D$$

$$= \frac{1}{Nh^D} \sum_{i=1}^{N} \int_{x_1 \in \mathbb{R}} \cdots \int_{x_D \in \mathbb{R}} K\left(\frac{\boldsymbol{x} - \boldsymbol{x}_i}{h}\right) \mathrm{d}x_1 \ldots \mathrm{d}x_D$$

$$= \frac{1}{Nh^D} \sum_{i=1}^{N} \int_{u_1 = -\infty}^{+\infty} \cdots \int_{u_D = -\infty}^{+\infty} K(\boldsymbol{u})(h\mathrm{d}u_1) \ldots (h\mathrm{d}u_D)$$

$$= \frac{1}{Nh^D} \sum_{i=1}^{N} h^D = \frac{1}{Nh^D} Nh^D = 1$$

Question 14:
Instead of entropy in decision trees, one can use the Gini index or the misclassification error. You have a data set with 400 examples from o class (i.e., 400°), 400 examples from $\square$ class (i.e., $400^{\square}$), and 400 examples from $\triangle$ class (i.e., $400^{\triangle}$). Suppose that you have two possible splits for a decision node m. The first splitting choice results in L_m: $(300^{\circ}, 200^{\square}, 100^{\triangle})$ and R_m: $(100^{\circ}, 200^{\square}, 300^{\triangle})$. The second splitting choice results in L_m: $(400^{\circ}, 200^{\square}, 0^{\triangle})$ and R_m: $(0^{\circ}, 200^{\square}, 400^{\triangle})$.

- Gini index with $K > 2$ classes: $\sum_{c=1}^{K} \sum_{d>c} 2p_c p_d$
- Misclassification error: $1 - \max(p_1, p_2, \ldots, p_K)$

(a) Calculate the impurity after these two choices of splitting using the Gini index and the misclassification error?

(b) Which of these two choices of splitting would be preferred and why?

(a)

$$\text{Gini}(1^{st}\text{ split}) = \frac{6}{12}\left(2\frac{3}{6}\frac{2}{6} + 2\frac{3}{6}\frac{1}{6} + 2\frac{2}{6}\frac{1}{6}\right) + \frac{6}{12}\left(2\frac{1}{6}\frac{2}{6} + 2\frac{1}{6}\frac{3}{6} + 2\frac{2}{6}\frac{3}{6}\right) = \frac{11}{18}$$

$$\text{Gini}(2^{nd}\text{ split}) = \frac{6}{12}\left(2\frac{4}{6}\frac{2}{6} + 2\frac{4}{6}\frac{0}{6} + 2\frac{2}{6}\frac{0}{6}\right) + \frac{6}{12}\left(2\frac{0}{6}\frac{2}{6} + 2\frac{0}{6}\frac{4}{6} + 2\frac{2}{6}\frac{4}{6}\right) = \frac{8}{18}$$

$$\text{Error}(1^{st}\text{ split}) = \frac{6}{12}\left[1 - \max\left(\frac{3}{6}, \frac{2}{6}, \frac{1}{6}\right)\right] + \frac{6}{12}\left[1 - \max\left(\frac{1}{6}, \frac{2}{6}, \frac{3}{6}\right)\right] = \frac{1}{2}$$

$$\text{Error}(2^{nd}\text{ split}) = \frac{6}{12}\left[1 - \max\left(\frac{4}{6}, \frac{2}{6}, \frac{0}{6}\right)\right] + \frac{6}{12}\left[1 - \max\left(\frac{0}{6}, \frac{2}{6}, \frac{4}{6}\right)\right] = \frac{1}{3}$$

(b) Both the Gini index and misclassification error prefer the second split.